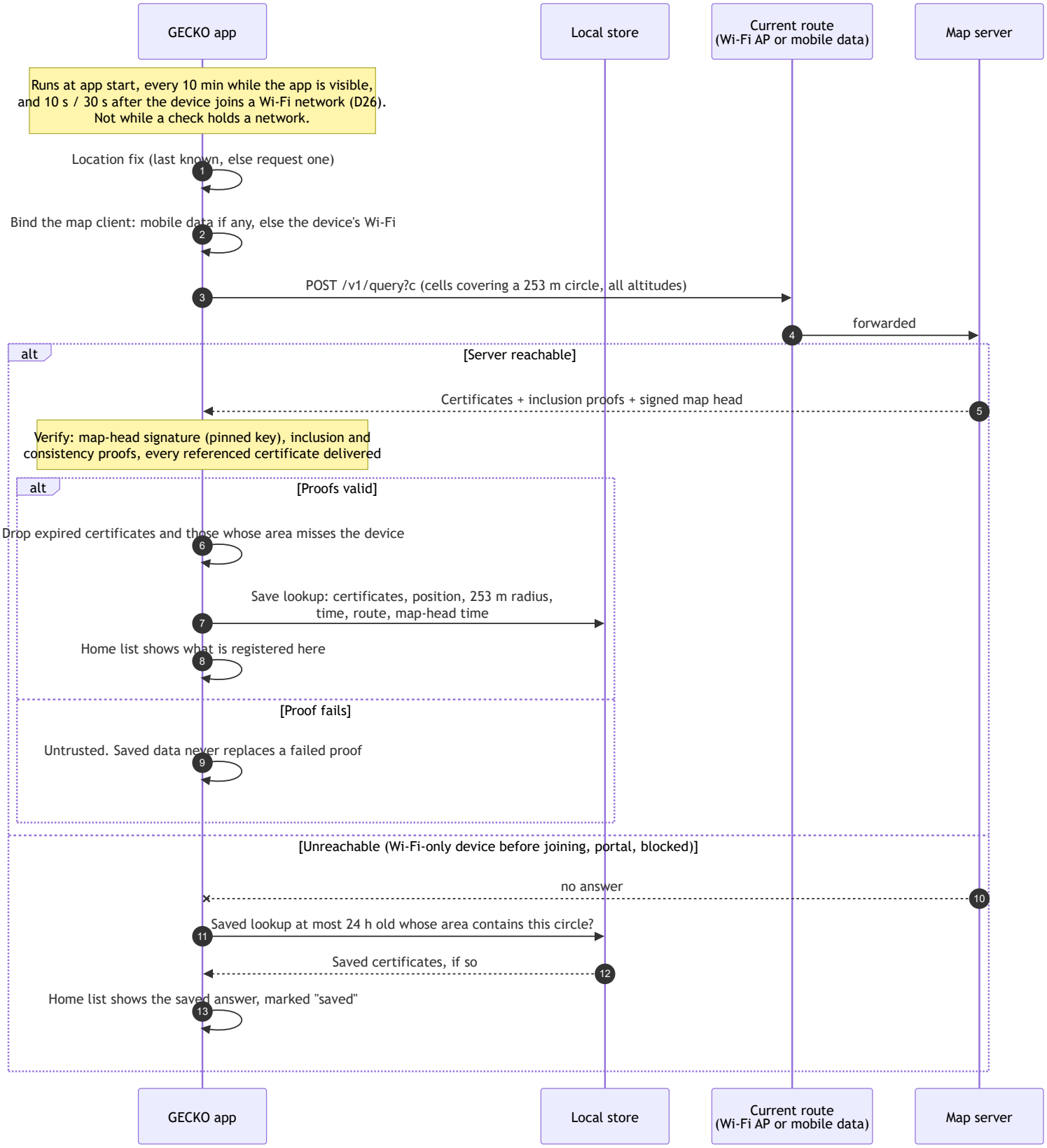

GECKO app
Local store
Current route
(Wi-Fi AP or mobile data)
Map server
Runs at app start, every 10 min while the app is visible,
and 10 s / 30 s after the device joins a Wi-Fi network (D26).
Not while a check holds a network.
Location fix (last known, else request one)
1
Bind the map client: mobile data if any, else the device's Wi-Fi
2
POST /v1/query?c (cells covering a 253 m circle, all altitudes)
3
forwarded
4
alt
[Server reachable]
Certificates + inclusion proofs + signed map head
5
Verify: map-head signature (pinned key), inclusion and
consistency proofs, every referenced certificate delivered
alt
[Proofs valid]
Drop expired certificates and those whose area misses the device
6
Save lookup: certificates, position, 253 m radius,
time, route, map-head time
7
Home list shows what is registered here
8
[Proof fails]
Untrusted. Saved data never replaces a failed proof
9
[Unreachable (Wi-Fi-only device before joining, portal, blocked)]
no answer
10
Saved lookup at most 24 h old whose area contains this circle?
11
Saved certificates, if so
12
Home list shows the saved answer, marked "saved"
13
GECKO app
Local store
Current route
(Wi-Fi AP or mobile data)
Map server